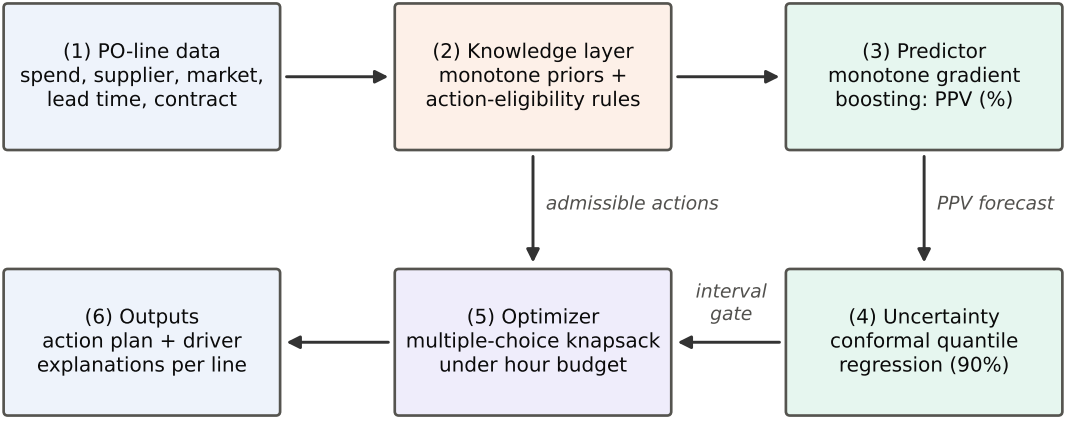
(1) PO-line data
spend, supplier, market,
lead time, contract
(2) Knowledge layer
monotone priors +
action-eligibility rules
(3) Predictor
monotone gradient
boosting: PPV (%)
admissible actions
PPV forecast
(6) Outputs
action plan + driver
explanations per line
(5) Optimizer
multiple-choice knapsack
under hour budget
interval
gate
(4) Uncertainty
conformal quantile
regression (90%)